RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Go, change the world

Academic year 2024-2025 (Odd Sem)
DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

Date	28-01-2025	Maximum Marks	10+
Course Code	CS354TA	Duration	120
Sem-V	Improvement Test	Staff: HKK|ASP|SMS|MRA	

THEORY OF COMPUTATION
(Common to CSE & ISE)

PART-A

		Marks	BT
1.	Design the TM that performs 2's complement.	2	L3
2.	Mention the Transition function δ for TM with stay option.	1	L1
3.	Difference between recursively enumerable language and recursive language.	2	L2
4.	Write the TM transition diagram for the given regular expression (0+1)*001(0+1)*.	2	L3
5.	What language does the following Unrestricted Grammar derive? $S \rightarrow S_1B,\ S_1 \rightarrow As_1b,\ Bb \rightarrow bbbB,\ As_1b \rightarrow aa,\ B \rightarrow \lambda$	2	L3
6.	Show that every regular languages are also recursively enumerable languages.	1	L1

PART-B

1.	Define Turing Machine, Language acceptance by Turing Machine. Design Turing Machine to accept $L = \{WW^R : W \in \{a, b, c\}^*$. Using Instantaneous Descriptions show that the string babccbab is accepted by TM.	10	L4
2a.	Define Linear Bounded Automata, Language acceptance by Linear Bounded Automata. Design Linear Bounded Automata to accept $L = \{ w \mid w \in \{a, b\}^* \text{ and } N_a(w)=N_b(w)\}$. Using Instantaneous Descriptions show that the string babaab is accepted by LBA.	6	L4
2b.	With the help of Chomsky hierarchy explain the relationship among families of languages.	4	L2
3a.	Design a Turing Machines (TM) to compute the function $f(x, y)=xy$ where $x, y \in \{a,b,c\}^*$. Trace the machine for x=bcab and y= abca.	6	L1
3b.	Explain with an example the working of Multitape Turing Machine. Prove that for the given Multitape Turing Machine there exists an equivalent single tape standard Turing Machine.	4	L1

4.	Define PCP.Solve the PCP given below.	6	L

	List A	List B
i	wi	Xi
1	10	101
2	01	100
3	0	10
4	100	0
5	1	010

4b.	Prove (i) If L1 and L2 are both recursively enumerable languages over Σ, then L1 ∪ L2 and L1 ∩ L2 are also recursively enumerable. (ii) If L1 and L2 are both recursive languages over Σ, then L1 ∪ L2 and L1 ∩ L2 are also recursive.	4	L2
5a.	Define Unrestricted Grammars. Write Unrestricted grammar to generate language $L = \{ w \mid w \in \{a, b, c\}^* \text{ and } N_a(w)=N_b(w)=N_c(w)\}$. Give the derivation for the string bacbbcaac.	4	L3
5b.	Design a Turing Machines (TM) to compute the function f(x, y) where x, y ∈ {1}*. $f(x, y) = x - y$ if $x > y$ $= y-x$ if $x < y$ $=0$ if $x=y$	6	L4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
Marks Distribution	Max Marks	8	14	16	22	-	12	10	16	22

Course Outcomes:

1. Understand the fundamental concepts of theory of computations
2. Analyze the tools of finite automata to various fields of computer science.
3. Design solution model for complex problems, using the appropriate skills of automata theory for
4. Apply automata skills in situations that describe computation effectively and efficiently.

****